

Supplementary Material

Cost-Related Barriers to Care and Medications Among US Stroke Survivors, 2019–2025

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Reference numbers refer to the reference list of the main manuscript. Contents: Supplemental Methods; Figure S1; Tables S1–S12 (Table S12 is the STROBE checklist).

Supplementary Methods

Data sources and harmonization

Microdata came from the seven annual NHIS Sample Adult files for 2019–2025, the seven annual multiply imputed income files, and the 2020 partial Sample Adult file. Documentation comprised the annual Sample Adult codebooks, survey descriptions, and income imputation technical documents. The download location, size, and SHA-256 hash of each of the 36 files are recorded in the source manifest distributed with the analysis code.

A questionnaire crosswalk records the exact wording, response universe, and codebook page of each stroke and affordability question in every year; a covariate crosswalk does the same for education, insurance, clinical, smoking, and disability variables. Question wording and universes were substantively identical across 2019–2025 after normalizing capitalization in the 2019 prescription-use item. Stroke history was defined as a reported prior clinician diagnosis; neither subtype nor recency was inferred.

Annual estimates retain the full 2020 Sample Adult file, including follow-back respondents, with the annual weight WTFA_A. Pooled estimates retain only the 2020 records that link to the partial file and use the partial-sample weight WTSA_P, matching households on the partial-file identifier. All other years use WTFA_A. Pooled weights were divided by seven so that weighted totals represent an average annual population; no total is presented as a count of unique survivors across years.

Survey designs were specified on the full adult files using the published stratum (PSTRAT) and primary sampling unit (PPSU) identifiers, which NCHS instructs users to combine across years without modification for 2019 onward. Domain estimation for stroke survivors retained the full design, so variance estimates account for the fact that the stroke domain is a subset of each sampling unit.

Outcome classification

Table S10. Outcome definitions and NHIS source variables

Measure	NHIS variables	Universe
Stroke history	STREV_A	All sample adults

Measure	NHIS variables	Universe
Delayed medical care	MEDDL12M_A	All sample adults
Forgone medical care	MEDNG12M_A	All sample adults
Forgone needed prescriptions	RXDG12M_A	All sample adults
Prescription use	RX12M_A	All sample adults
Dose skipping, dose reduction, delayed filling	RXSK12M_A; RXLS12M_A; RXDL12M_A	RX12M_A=1
Any affordability barrier	Any applicable positive among six items	Known applicability and response rules described below

Response codes 1 and 2 indicate "yes" and "no." Refused, not ascertained, don't know, and missing responses were never recoded as "no." Any applicable "yes" defined a positive composite even if another component was unknown. An all-negative composite required a known response to every applicable item. Prescription nonuse (RX12M_A=2) is an explicit branch that leaves the three medication-use items structurally blank; if prescription use was unknown and all universal items were negative, the primary outcome was classified as unknown. The medication-underuse denominator includes only respondents who reported prescription use and had a classifiable underuse outcome.

Covariate definitions

Table S11. Covariate definitions and NHIS source variables

Domain	Variables	Analysis categories
Age	AGEP_A	18–85; 85 is top-coded; natural spline or 18–64/65+
Sex	SEX_A	Male; female
Race/Hispanic origin	HISPALLP_A	Seven published categories
Region	REGION	Northeast; Midwest; South; West
Education	EDUC_A; EDUCP_A	Four harmonized attainment groups
Income	RATCAT_A; IMPNUM or IMPNUM_A	<100%; 100–199%; 200–399%; ≥400% FPL
Insurance	NOTCOV_A and seven coverage indicators	Private; public without private; military only; uninsured
Clinical history	HYPEV_A; DIBEV_A; CHDEV_A	Yes/no hypertension, diabetes, coronary disease
Smoking	SMKCIGST_A	Never; former; current; indeterminate missing
Self-rated health	PHSTAT_A	Fair/poor vs good/very good/excellent
Disability	DISAB3_A	Published Washington Group disability recode

For the individual coverage indicators, codes 1 and 2 both indicate coverage and code 3 indicates no coverage; the uninsured variable NOTCOV_A codes 1 as not covered and 2 as covered. The insurance hierarchy assigned private coverage first among covered respondents, then public coverage without private coverage, then military coverage only when every other listed coverage type was explicitly absent. Ambiguous coverage patterns were set to missing. Private coverage may coexist with public or military coverage, and public-without-private coverage may coexist with military coverage; military-only status is therefore deliberately narrow.

Income used the categorical poverty-ratio variable RATCAT_A rather than the rounded continuous ratio. Categories 1–3 correspond to below 100% of the federal poverty level (FPL), 4–7 to 100%–199%, 8–11 to 200%–399%, and 12–14 to 400% or higher. The imputation index is named IMPNUM in 2019–2020 and IMPNUM_A thereafter. All ten annual imputations were linked by year, household, and index, with assertions of unique keys and complete matching. In

descriptive tables the income point estimate is the mean of the ten completed-data weighted percentages, and unweighted income counts are averages over imputations, which is why they may be fractional.

Estimands and models

The primary descriptive estimand is the survey-weighted prevalence of any cost-related barrier among noninstitutionalized adults reporting prior stroke with a classifiable outcome. Adjusted prevalence ratios condition on the stated covariate set and are descriptive associations, not causal effects, incidence ratios, or recurrence estimates. Model 1 adjusts for categorical year, an age spline, sex, race and Hispanic origin, and region. Model 2 adds education, income, and insurance. Model 3 adds hypertension, diabetes, coronary heart disease, smoking, self-rated health, and disability. All three models use the same 6,777 records. The age-group model sequence substitutes a working-age indicator for the age spline.

For each model, coefficients were averaged over the ten imputations. The total covariance equals the mean within-imputation survey covariance plus 1.1 times the between-imputation covariance of the coefficients. Confidence intervals use the finite-sample degrees of freedom returned by `mitools::MIcombine`, supplied with the minimum residual design degrees of freedom across the ten fits. Poisson coefficients were exponentiated to obtain prevalence ratios. This procedure propagates income-imputation uncertainty; it does not impute missing outcomes or other covariates.

For standardized annual prevalence, the fitted logistic model assigned each survivor to each survey year in turn while holding other covariates fixed, and the predicted probabilities were averaged with the pooled complete-case weights. The linearized influence function includes the regression-coefficient influence multiplied by the prediction gradient and the weighted deviation of each predicted probability from its standardized mean, so the covariance incorporates uncertainty in the reference distribution and its covariance with model estimation. Imputation-specific margins and their covariance were pooled on the logit scale; the 2025-minus-2019 absolute difference was pooled on the probability scale.

The bounded logistic sensitivity applied the same standardization to income, insurance, disability, and the age-group model, assigning the full analytic sample to the comparison and reference categories in turn and forming their prevalence ratio. These marginal estimands differ from conditional modified Poisson prevalence ratios and can involve extrapolation to sparsely supported covariate combinations; agreement between the two approaches is a model-sensitivity observation rather than proof of correct specification.

The categorical-year and year-by-age-group tests are pooled Wald F statistics based on the selected coefficients and their Rubin total covariance, with denominator degrees of freedom equal to the minimum scalar pooled degrees of freedom among those coefficients. This is an approximate multivariate multiple-imputation test; because the missing-information fractions for

the year terms were small, it provides a practical summary, and the estimates and intervals remain the primary description of annual variation.

Confidence intervals and precision screen

Annual observed confidence intervals are design-based logit intervals. For pooled component estimates, an effective sample size was calculated from the prevalence and its Taylor variance, capped at the observed sample size and adjusted by the squared ratio of the 0.975 normal quantile to the corresponding design-based t quantile; beta quantiles based on that effective sample size provide bounded intervals. Subgroup intervals were pooled across imputations on the logit scale.

Following NCHS data presentation standards,[11] a proportion was flagged when its unweighted or effective sample size was below 30, its confidence-interval width exceeded 0.30, or its width exceeded 0.05 and was more than 130% of the smaller of the estimate and its complement; design degrees of freedom below eight prompted review. All main annual, component, and reported subgroup estimates passed. The unknown-insurance subgroup (29 records) failed the screen and is not reported.

Sensitivity analyses and verification

Component-outcome models used the primary-model complete-covariate population restricted to observed values of the component, so their denominators differ from the broader component prevalence denominators. The no-disability model omits the disability term. The working-age model retains a continuous age spline within ages 18–64 years. The no-2020 model removes 2020 records and the corresponding year level. The missing-category analysis retains all 7,104 observed primary outcomes and includes explicit unknown levels for incomplete nonincome covariates.

All 160 modified Poisson fits and all 20 bounded logistic fits converged. The fully adjusted modified Poisson model produced 14–16 fitted values above one per imputation, with a maximum of approximately 1.9; these are reported as diagnostics and were not interpreted as individual probabilities. Independent verification in Python reconstructed 12 annual and pooled primary estimates and their Taylor standard errors directly from the source files, reproduced the first-imputation fully adjusted model from the exported design matrix (maximum absolute coefficient discrepancy below 3×10^{-13} ; maximum covariance discrepancy below 3×10^{-9}), and checked Rubin-pooled means and standard errors for all 16 model specifications.

Analytic decisions

Employment status was not included because it overlaps strongly with age and disability and would have required additional harmonization across years. A year-by-insurance interaction was not fit because the military-only group contributed only 109 observed records and 17 events across the seven years. No data-driven significance criterion was used to select the final model. The ten records with unknown age were excluded rather than imputed.

Supplementary Figure

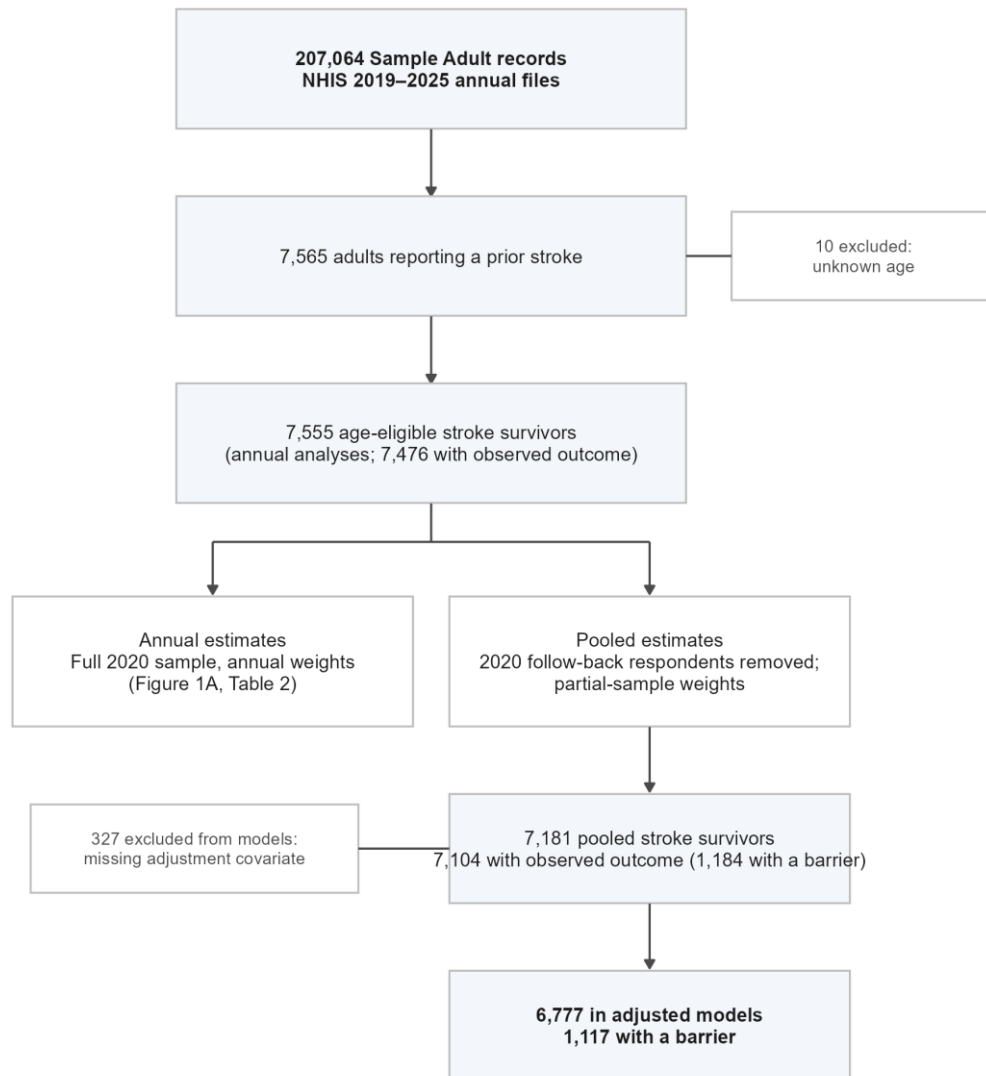


Figure S1. Selection of the annual and pooled analytic samples. Annual and pooled samples apply different inclusion and weighting rules for 2020. The 7,555 annual stroke records include the full 2020 sample; the 7,181 pooled stroke records exclude the 2020 follow-back respondents and use the partial-sample weights. Adjusted models additionally require an observed outcome and complete nonincome covariates. Counts are records, not unique persons across years.

Supplementary Tables

Table S1. Annual and pooled sample counts by survey year

Year	Annual stroke	Annual outcome observed	Pooled stroke	Pooled outcome observed	Pooled events
2019	1201	1191	1201	1191	231
2020	1059	1051	685	679	105
2021	1010	1001	1010	1001	179
2022	1008	999	1008	999	147
2023	1131	1116	1131	1116	169
2024	1244	1229	1244	1229	215
2025	902	889	902	889	138

Annual stroke records total 7,555; pooled stroke records total 7,181. The difference comes from excluding 2020 followback records in pooled analyses. These are unweighted counts.

Table S2. Missing nonincome covariates in the pooled stroke-survivor sample

Variable	Missing n	Missing %
any_barrier	77	1.07
age	0	0.00
sex	0	0.00
race	0	0.00
region	0	0.00
education	35	0.49
insurance	34	0.47
hypertension	9	0.13
diabetes	10	0.14
coronary	68	0.95
fairpoor	11	0.15
smoking	253	3.52
disability	0	0.00

Denominator is 7,181 stroke records. Missingness overlaps across variables; the counts must not be summed to derive complete cases. Among records with observed primary outcomes, 327 had at least one missing adjustment variable.

Table S3. Complete characteristics of the pooled sample, overall and by cost-related barrier status

Characteristic	Overall (N=7,104)	No barrier (n=5,920)	Any barrier (n=1,184)
Age 65 years or older	4,763 (59.0)	4,237 (64.0)	526 (35.9)
Age 18–64 years	2,341 (41.0)	1,683 (36.0)	658 (64.1)
Male	3,269 (48.5)	2,777 (49.7)	492 (43.3)
Female	3,835 (51.5)	3,143 (50.3)	692 (56.7)
Hispanic, any race	599 (11.5)	478 (11.2)	121 (12.8)
Non-Hispanic White only	5,052 (66.6)	4,279 (67.6)	773 (61.7)

Characteristic	Overall (N=7,104)	No barrier (n=5,920)	Any barrier (n=1,184)
Non-Hispanic Black only	1,068 (15.8)	852 (14.9)	216 (19.7)
Non-Hispanic Asian only	171 (3.3)	146 (3.5)	25 (2.4)
Non-Hispanic AIAN only	77 (0.9)	66 (0.9)	11 (0.8)
Non-Hispanic AIAN with another race	91 (1.3)	63 (1.1)	28 (1.8)
Other single or multiple races	46 (0.7)	36 (0.6)	10 (0.8)
Northeast	1,032 (15.6)	901 (16.4)	131 (11.8)
Midwest	1,681 (22.0)	1,405 (22.3)	276 (20.6)
South	2,942 (43.1)	2,362 (41.0)	580 (52.7)
West	1,449 (19.4)	1,252 (20.3)	197 (15.0)
Bachelor's degree or higher	1,670 (20.1)	1,437 (20.6)	233 (17.4)
Some college or associate	2,149 (28.7)	1,743 (28.2)	406 (30.9)
High school or GED	2,202 (32.3)	1,848 (32.6)	354 (30.8)
Less than high school	1,050 (18.3)	866 (18.0)	184 (19.7)
Education unknown	33 (0.6)	26 (0.5)	7 (1.2)
Income $\geq$ 400% FPL	1,681.2 (24.2)	1,559.9 (27.0)	121.3 (11.0)
Income 200–399% FPL	2,133.3 (31.0)	1,786.9 (31.0)	346.4 (31.2)
Income 100–199% FPL	1,986.9 (27.5)	1,549.6 (25.6)	437.3 (36.0)
Income <100% FPL	1,302.6 (17.4)	1,023.6 (16.4)	279.0 (21.8)
Private, with or without public coverage	2,888 (40.3)	2,465 (40.9)	423 (37.2)
Public without private	3,871 (53.6)	3,263 (54.9)	608 (47.4)
Military only	109 (1.6)	92 (1.6)	17 (1.7)
Uninsured	207 (4.0)	80 (2.0)	127 (13.2)
Insurance unknown	29 (0.5)	20 (0.5)	9 (0.6)
No hypertension	1,747 (25.7)	1,455 (25.3)	292 (27.8)
Hypertension	5,349 (74.2)	4,458 (74.6)	891 (72.1)
Hypertension unknown	8 (0.1)	7 (0.1)	1 (0.1)
No diabetes	5,136 (71.3)	4,330 (72.3)	806 (66.9)
Diabetes	1,960 (28.6)	1,583 (27.6)	377 (33.0)
Diabetes unknown	8 (0.1)	7 (0.1)	1 (0.0)
No coronary heart disease	5,212 (74.0)	4,369 (74.8)	843 (70.8)
Coronary heart disease	1,828 (25.2)	1,500 (24.5)	328 (28.4)
Coronary heart disease unknown	64 (0.7)	51 (0.7)	13 (0.8)
Never smoked	3,196 (46.0)	2,710 (46.6)	486 (43.5)
Former smoker	2,592 (34.0)	2,240 (35.5)	352 (27.2)
Current smoker	1,132 (16.8)	821 (14.7)	311 (26.3)
Smoking unknown	184 (3.2)	149 (3.2)	35 (3.0)
Good or better self-rated health	3,607 (48.0)	3,201 (51.6)	406 (31.7)
Fair or poor self-rated health	3,486 (51.9)	2,712 (48.3)	774 (68.1)
Self-rated health unknown	11 (0.1)	7 (0.1)	4 (0.2)
Without disability	4,384 (61.2)	3,742 (62.3)	642 (56.3)
With disability	2,720 (38.8)	2,178 (37.7)	542 (43.7)

Values are unweighted number (survey-weighted column percentage). Income counts are averaged across the ten NCHS imputations and may be fractional. Unknown values are retained in denominators; percentages within a variable may not sum to

100 because of rounding. FPL indicates federal poverty level; NHIS, National Health Interview Survey. AIAN, American Indian or Alaska Native. The race/Hispanic-origin categories follow the published public-use recode.

Table S4. Pooled prevalence of the composite outcome and each component barrier, 2019–2025

Outcome	No. with observed outcome	No. with barrier	Prevalence, % (95% CI)
Any affordability barrier	7104	1184	17.9 (16.8–19.1)
Three universal questions	7106	1012	15.5 (14.3–16.6)
Delayed medical care	7116	554	8.5 (7.7–9.4)
Forgone medical care	7115	541	8.4 (7.6–9.3)
Forgone needed prescriptions	7103	647	10.0 (9.1–11.0)
Medication underuse among users	6754	635	10.1 (9.2–11.1)

Components overlap. Medication-underuse estimates use prescription users only. Each prevalence denominator includes all stroke records with a classifiable corresponding outcome; it is not restricted to complete model covariates. All displayed estimates pass the implemented precision screen.

Table S5. Fully adjusted (Model 3) prevalence ratios for all covariates

Characteristic	PR (95% CI)	P value
Year 2020 vs 2019	0.89 (0.69–1.14)	0.351
Year 2021 vs 2019	0.96 (0.78–1.19)	0.702
Year 2022 vs 2019	0.84 (0.67–1.04)	0.112
Year 2023 vs 2019	0.85 (0.69–1.05)	0.140
Year 2024 vs 2019	1.06 (0.87–1.29)	0.569
Year 2025 vs 2019	0.89 (0.71–1.10)	0.274
Female vs male	1.27 (1.12–1.43)	<0.001
Non-Hispanic White only vs Hispanic	1.20 (0.95–1.51)	0.124
Non-Hispanic Black only vs Hispanic	1.14 (0.88–1.48)	0.323
Non-Hispanic Asian only vs Hispanic	0.78 (0.45–1.35)	0.367
Non-Hispanic AIAN only vs Hispanic	0.82 (0.42–1.61)	0.570
Non-Hispanic AIAN with another race vs Hispanic	1.06 (0.69–1.63)	0.800
Other single or multiple races vs Hispanic	1.26 (0.71–2.25)	0.431
Midwest vs Northeast	1.21 (0.97–1.50)	0.098
South vs Northeast	1.42 (1.15–1.75)	<0.001
West vs Northeast	1.24 (0.96–1.61)	0.095
Some college/associate vs bachelor's or higher	0.89 (0.75–1.06)	0.202
High school/GED vs bachelor's or higher	0.75 (0.62–0.89)	0.001
Less than high school vs bachelor's or higher	0.80 (0.64–0.99)	0.038
Income 200–399% vs ≥400% FPL	2.07 (1.65–2.61)	<0.001
Income 100–199% vs ≥400% FPL	2.50 (1.94–3.21)	<0.001
Income <100% vs ≥400% FPL	2.00 (1.52–2.63)	<0.001
Public without private vs private	0.72 (0.63–0.83)	<0.001
Military only vs private	0.81 (0.52–1.28)	0.372
Uninsured vs private	1.79 (1.49–2.16)	<0.001

Characteristic	PR (95% CI)	P value
Hypertension vs none	0.99 (0.86–1.14)	0.863
Diabetes vs none	1.12 (0.97–1.30)	0.126
Coronary heart disease vs none	1.37 (1.18–1.58)	<0.001
Former vs never smoking	0.98 (0.85–1.14)	0.820
Current vs never smoking	1.21 (1.04–1.42)	0.015
Fair/poor vs good or better health	1.58 (1.37–1.82)	<0.001
Disability vs none	1.19 (1.05–1.35)	0.006

The model includes an age spline with 3 degrees of freedom. Spline basis coefficients and the intercept are retained in model_coefficients.csv; they do not individually represent clinical group contrasts. No multiplicity correction was applied. Sparse race and military-coverage contrasts require caution.

Table S6. Selected prevalence ratios in sensitivity analyses

Analysis	No.	Income 100–199% vs ≥400% FPL	Uninsured vs private	Disability vs none
Exclude 2020	6,124	2.59 (2.00–3.36)	1.82 (1.49–2.21)	1.18 (1.04–1.34)
Age 18–64 only	2,231	2.33 (1.67–3.25)	1.85 (1.51–2.27)	1.08 (0.92–1.27)
No disability	4,209	2.50 (1.79–3.47)	1.80 (1.42–2.28)	Not applicable
Universal composite	6,776	2.45 (1.87–3.22)	2.09 (1.70–2.57)	1.26 (1.10–1.45)
Delayed medical care	6,774	2.58 (1.76–3.79)	2.69 (1.98–3.66)	1.35 (1.10–1.65)
Forgone medical care	6,776	2.46 (1.66–3.65)	2.99 (2.23–4.00)	1.40 (1.15–1.71)
Forgone prescriptions	6,775	2.52 (1.75–3.65)	2.09 (1.58–2.77)	1.24 (1.04–1.49)
Medication underuse	6,449	2.85 (2.01–4.03)	1.74 (1.32–2.31)	1.23 (1.03–1.48)
Unknown-category diagnostic	7,104	2.45 (1.91–3.14)	1.85 (1.55–2.21)	1.16 (1.03–1.31)

Values are PRs (95% CIs) from the corresponding fully adjusted model. No-disability models omit the disability term. The complete coefficient sets, including all income categories, are supplied as CSV. Working-age disability estimates are compatible with no association. Outcome-specific model denominators differ from Table S4 because they require complete adjustment variables.

Table S7. Standardized prevalence ratios from bounded logistic models

Contrast	Marginal PR (95% CI)	Reference %	Comparison %
Income 200–399% FPL vs Income ≥400% FPL	1.99 (1.59–2.49)	9.6	19.0
Income 100–199% FPL vs Income ≥400% FPL	2.41 (1.88–3.09)	9.6	23.0
Income <100% FPL vs Income ≥400% FPL	1.91 (1.45–2.51)	9.6	18.3
Public without private vs Private, with or without public coverage	0.72 (0.63–0.83)	20.1	14.5
Military only vs Private, with or without public coverage	0.83 (0.52–1.32)	20.1	16.7
Uninsured vs Private, with or without public coverage	1.99 (1.63–2.43)	20.1	40.0
With disability vs Without disability	1.18 (1.04–1.33)	16.7	19.6
Age 18–64 years vs Age 65 years or older	2.13 (1.86–2.44)	12.0	25.5

Logistic predictions remain between zero and one. Marginal PRs are pooled on the log scale; displayed prevalences are arithmetic means of imputation-specific margins, so their displayed ratio may differ slightly from the pooled PR. The age contrast uses the age-group model. These standardized estimates have a different estimand from conditional modified Poisson PRs.

Table S8. Unadjusted pooled prevalence of any cost-related barrier in principal subgroups

Subgroup	No. with observed outcome	Prevalence, % (95% CI)
Age 65 years or older	4,763	10.9 (9.8–12.0)
Age 18–64 years	2,341	28.0 (25.9–30.2)
Income $\geq$ 400% FPL	1681.2	8.2 (6.7–10.0)
Income 200–399% FPL	2133.3	18.0 (16.0–20.3)
Income 100–199% FPL	1986.9	23.5 (21.1–26.1)
Income <100% FPL	1302.6	22.5 (19.6–25.6)
Private, with or without public coverage	2,888	16.6 (14.9–18.4)
Public without private	3,871	15.8 (14.5–17.3)
Military only	109	18.2 (11.1–28.4)
Uninsured	207	58.7 (50.7–66.3)
Without disability	4,384	16.5 (15.1–17.9)
With disability	2,720	20.2 (18.4–22.1)

Income counts are averages across ten imputations. The unknown-insurance subgroup (29 records) failed the precision screen, and its prevalence is not displayed. These are unadjusted subgroup estimates.

Table S9. Convergence and fitted-value diagnostics for the modified Poisson models

Model	n	Events	Fitted >1 per imputation	Largest fitted value	Convergence
M1	6777	1117	0–0	0.517	All converged
M2	6777	1117	3–4	1.190	All converged
M3	6777	1117	14–16	1.943	All converged
Age_M1	6777	1117	0–0	0.466	All converged
Age_M2	6777	1117	3–4	1.106	All converged
Age_M3	6777	1117	15–17	1.912	All converged
AgeInteraction	6777	1117	17–19	1.888	All converged
No2020	6124	1016	12–16	1.943	All converged
WorkingAge	2231	620	14–15	2.127	All converged
NoDisability	4209	607	5–6	1.802	All converged
Universal	6776	955	13–17	1.972	All converged
Delayed	6774	523	3–5	1.518	All converged
ForgoneCare	6776	509	4–8	1.651	All converged
ForgoneRx	6775	610	4–6	1.488	All converged
RxUnderuse	6449	598	5–6	1.709	All converged
MissingCategory	7104	1184	16–19	1.891	All converged

Ten fits per specification. Fitted log-link values above one are not interpreted as individual probabilities. The bounded logistic sensitivity supports the major associations; it does not establish perfect specification. Model labels correspond to the accompanying script and coefficient CSV.

STROBE Checklist

Table S12. STROBE checklist for cross-sectional studies

Item	Location in manuscript	Content
1	Title and Abstract	Design, population, period and principal quantitative findings.
2–3	Introduction	Background, prior studies and objectives stated.
4–5	Methods / Study design	Repeated cross-sectional design, public NHIS setting and 2019–2025 period.
6	Methods / Population; Figure S1	Eligibility, stroke definition, age exclusion, annual vs pooled selection.
7–8	Methods / Outcomes and covariates	Definitions, universes and cross-year source verification; crosswalk files included.
9	Discussion / Limitations	Self-report, nonresponse, institutional exclusion, confounding and missingness addressed.
10	Methods / Population	All eligible records in the selected years; no formal power-based sample selection.
11	Methods / Covariates	Age spline and categorical income/education definitions given.
12	Methods / Analysis; Supplementary Methods	Survey weighting, MI, missingness, interaction and sensitivity procedures.
13	Results / Sample; Table S1; Figure S1	Counts and reasons for exclusions; different annual and pooled paths.
14	Table 1; Tables S2–S3	Characteristics and missingness; income counts averaged across imputations.
15	Table 2; Tables S4 and S8	Outcome counts and annual, component and subgroup estimates.
16	Table 3; Table S5; Results	Adjusted PRs, covariate sets, precision and absolute annual difference.
17	Sensitivity Results; Tables S6–S9	Sensitivity estimates and model diagnostics.
18–21	Discussion and Conclusions	Key results, limitations, cautious interpretation and generalizability.
22	Declarations	No external funding is reported.